Lecture Notes on

Introduction to Astrophysics

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These notes summarize the first Astrophysics Club meeting. The goal is to outline the purpose of the club and introduce fundamental concepts about the scale and structure of the universe.

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1 Purpose of the Club

The Astrophysics Club was created to:

- Explore astrophysics topics not typically covered in Fremont courses.
- Organize telescope observations and guest lectures.
- Discuss the physics governing the universe.
- Encourage critical thinking through competitions and projects.
- Build curiosity about large-scale structure and cosmic phenomena.

Leadership:

- President: Abir Mehta
- Vice President: Dhruv Lagu

2 Astrophysics Competitions

For students interested in deeper challenge:

- **IAAC** — International Astronomy and Astrophysics Competition.
- **USAAAO** — U.S. Astronomy and Astrophysics Olympiad.
- **USAAO** — U.S. Astronomy Olympiad.

Competitions emphasize:

- Problem solving
- Conceptual understanding
- Quantitative reasoning
- Application of physics to astronomy

3 Core Learning Objectives

Topics we aim to study include:

- Scale of the universe
- Celestial motion
- Light and electromagnetic radiation
- Stars and stellar evolution
- Galaxies

- Cosmology
- Exoplanets
- Relativity
- Black holes
- Telescopes and instrumentation
- History of astrophysics
- Current research frontiers

Underlying theme:

- Everything in astrophysics ultimately connects back to physics.
- Gravity, electromagnetism, thermodynamics, and nuclear physics govern cosmic behavior.

4 Scale of the Universe

Astrophysics is fundamentally about scale.

Key question:

- How large is the universe?
- How far apart are objects?
- How long do cosmic processes take?

Understanding scale is essential before studying advanced topics.

5 Distance Units in Astronomy

5.1 Light-Year

- A light-year is the distance light travels in one year.
- Speed of light:

$$c = 299{,}792{,}458 \text{ m/s}.$$

- Since light travels extremely fast, a light-year measures enormous distances.

Important clarification:

- A light-year is a unit of *distance*, not time.

5.2 Astronomical Unit (AU)

- 1 AU is the average distance between Earth and the Sun.
- Approximately 93 million miles.
- Useful for describing distances inside planetary systems.

5.3 Parsec

- 1 parsec $\approx$ 3.26 light-years.
- Common in professional astronomy.
- Derived from parallax measurements.

6 Cosmic Structure Hierarchy

From smallest to largest:

- Moon
- Planet
- Star
- Planetary system (e.g., Solar System)
- Star cluster
- Galaxy
- Galaxy group
- Galaxy cluster
- Supercluster
- Observable universe

6.1 Example: The Milky Way

- Diameter $\approx$ 100,000 light-years.
- Contains hundreds of billions of stars.
- Our Sun is located in one of its spiral arms.

7 Why Perspective Matters

Human intuition is built for everyday scales:

- meters
- kilometers
- seconds

Astrophysics operates on:

- billions of kilometers

- thousands of light-years
- billions of years

Developing intuition for these scales is the first step toward understanding astrophysical phenomena.

8 Looking Ahead

Future meetings will explore:

- Celestial motion
- Black holes
- Stellar evolution
- Relativity
- Cosmology

Astrophysics is only fun with people you know — invite friends and build the community.